Assignement-1

DSC624

Supply-Chain-Fraud-Analysis

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Shakira

1. Introduction

Supply chains are complex networks involving suppliers, manufacturers, distributors, and customers. With globalization, these networks have grown in scale and complexity, making them vulnerable to risks such as fraud, counterfeit goods, delayed shipments, and abnormal order behaviour. Detecting these anomalies early is critical for minimizing financial losses and maintaining customer trust.

Traditional anomaly detection methods often rely on numeric features such as transaction amounts or delivery times. While useful, these methods overlook the rich textual information embedded in order descriptions, customer notes, and shipping details. This project explores a hybrid approach that combines structured numeric features with unstructured textual data using large language model (LLM) embeddings. By doing so, we aim to capture both quantitative and semantic signals of fraud.

2. Problem Statement

The central problem addressed is **fraud detection in supply chain transactions**. Fraudulent activities can manifest in multiple ways:

- Orders with unusually high or low amounts.
- Suspicious shipping delays or unrealistic delivery times.
- Customer notes or descriptions that contain misleading or unusual language.

The dataset used is the **DataCo Supply Chain Dataset**, a large collection of transactional records including order details, customer information, and shipping logs. The objective is to build

models that can flag suspicious orders by analyzing both structured and unstructured data.

This project also emphasizes **reproducibility**: all code is contained in the repository, results are stored in the results/ folder, and the report documents methodology and findings.

3. Methodology

Baseline Models

Two baseline models were implemented:

- **Isolation Forest**: An unsupervised anomaly detection algorithm that isolates outliers by randomly partitioning data. It is particularly effective for high-dimensional datasets.
- **Gradient Boosting Classifier**: A supervised ensemble method that builds decision trees sequentially to minimize classification error. It is known for its robustness and ability to handle complex feature interactions.

Numeric features used included order amount, shipping days, delivery delays, and customer demographics. Preprocessing steps involved handling missing values, normalizing continuous variables, and encoding categorical features.

LLM Enhancement

To enrich the model, textual fields such as order descriptions and customer notes were embedded using the **SentenceTransformer** library. These embeddings capture semantic meaning, allowing the model to detect subtle fraud patterns that numeric features alone cannot reveal.

The embeddings were concatenated with numeric features to form a hybrid feature set. This hybrid dataset was then fed into

the Gradient Boosting classifier, creating a model that leverages both structured and unstructured information.

4. Experimental Setup

- **Environment**: Python, Jupyter Notebook, scikit-learn, SentenceTransformer.
- **Dataset location**: data/DataCoSupplyChainDataset.xlsx.
- **Results storage**: All metrics and plots are saved in the results/ folder (one level above the notebook).
- **Evaluation metrics**: Accuracy, Precision, Recall, F1 score, and ROC-AUC.
- **Workflow**:
    1. Load dataset.
    2. Train baseline models.
    3. Save baseline results (metrics + plots).
    4. Generate embeddings.
    5. Train hybrid model.
    6. Save hybrid results (metrics + plots).

This structured workflow ensures reproducibility and clarity.

5. Results

The baseline models provided a reasonable starting point. Gradient Boosting outperformed Isolation Forest in terms of precision and recall, but both struggled with subtle fraud patterns.

Once textual embeddings were added, the hybrid model showed a clear improvement across all metrics:

- **Metrics**: Stored in results/metrics.csv. The hybrid model achieved higher accuracy and F1 score compared to baselines.
- **Confusion Matrix**: (results/confusion_matrix.png) shows fewer false negatives in the hybrid model, meaning it caught more fraudulent cases.
- **ROC Curve**: (results/roc_curve.png) demonstrates a higher AUC for the hybrid model, indicating better discrimination between fraud and non-fraud.
- **Feature Importance**: (results/feature_importance.png) reveals that while numeric features like “order amount” remained strong predictors, textual embeddings contributed significantly to the model’s decision-making.

6. Analysis & Discussion

The hybrid approach demonstrates that fraud detection benefits from combining structured and unstructured data. Numeric features capture obvious anomalies, while textual embeddings uncover hidden patterns such as suspicious order descriptions or unusual customer notes.

Strengths:

- Improved accuracy and recall.
- Ability to detect subtle fraud patterns.
- Reproducible workflow with clear results storage.

Limitations:

- Dependency on external APIs for embeddings.

- Computational overhead due to large embeddings.
- Dataset size constraints — larger datasets may yield even better results.

Future Work:

- Fine-tuned LLMs for domain-specific fraud detection.
- Larger datasets for more robust training.
- Real-time anomaly detection pipelines integrated into supply chain management systems.

7. Conclusion

This project successfully integrates machine learning and LLM embeddings for supply chain fraud detection. The hybrid model outperforms traditional baselines, highlighting the importance of semantic understanding in anomaly detection.

With reproducible code, clear results stored in the results/ folder, and a structured methodology, this work provides a strong foundation for further research and practical deployment. The findings suggest that organizations can significantly enhance fraud detection by leveraging both numeric and textual data, ultimately leading to more secure and efficient supply chains.

8. References

- DataCo Supply Chain Dataset (Mendeley Data).
- scikit-learn documentation.
- SentenceTransformer library documentation.
- Gradient Boosting and Isolation Forest research papers.